

CENTRAL BANK OF SAMOA

CBS's Experience in Developing a GDP Nowcasting Framework

IMF STI Cohort Training in Nowcasting
Singapore, 2026

CBS Act 2015



1. Price Stability

Main Objective of CBS. Ensure that inflation is around its medium-term target of 3%

2. Financial Stability

Ensure that the financial system is sound and safe to provide financial services to the country.

3. Other (Govt) Objectives

Includes other objective such as

3.1 Economic Growth – support medium term growth of around 3% in the economy

3.2 Exchange Rates – ensure the exchange rate regime is conducive for exports but mitigates imported inflation

3.3 Foreign Reserves – ensure that the level of foreign reserves are adequate (above 4 months of imports) to cover for the country's import needs

Monetary Policy Framework

- CBS uses a monetary targeting regime with M2 as the intermediate target (although practice is a bit different)
- CBS uses fixed exchange rate regime (i.e. Pegged to a currency basket) (USD, AUD, NZD, EUR) – foreign exchange controls
- *(IMF MCM TA on Monetary Policy Framework to review this)*

Samoa's Stylized Facts

- Quarterly National Accounts (GDP release):
  - Quarterly
  - 90-day (3 months) lag
- Monetary policy decisions
  - Twice a year (June & December)
- Samoa is vulnerable to:
  - Natural Disasters
  - Tourism Fluctuations (Covid 19)
  - External Shocks i.e. Oil crisis, etc.

Why Nowcasting?

- Policymakers (Finance Minister, CBS & MOF) need a tool capable of assessing current economic activities before release of official GDP (3-month lag)
- With Nowcasting, CBS can:
  - Estimate current quarter GDP before official publication
  - Resolve issue (lag) in GDP preparation timeline and dissemination
  - Deliver accurate and prompt economic trend predictions; and
  - Conduct real-time economic analysis and formulate policy
- Thus, Nowcasting Framework supports:
  - Monetary Policy
  - Economic Surveillance
  - Communications

Development Phases

1. Indicator Selection
2. Model Evaluation
3. Back-testing
4. Documentation

1. Data/Indicator Selection

- Selecting predictors that can forecast real GDP effectively
- Include all accessible high-frequency indicators that could inform the dynamics of Samoa's real GDP;
  - E.g. tourism, remittances, agricultural production, external sector, fiscal, financial and monetary variables, El Niño indicator, etc.
- Indicator correlation/regression with GDP
- Availability vs Quality
- Determining/assessing which variables to include (next slide)

Data Processing and Econometric Work

Data Transformations

- Log transformation (Growth Rates)
- Quarterly conversions (Aggregation, Averaging or last observed value)

Statistical tests

- ADF unit-root testing
- Stationarity testing
- Seasonal adjustment (X-13 / Census X-13)
- Structural break dummies
  - COVID
  - Cyclones

Selection Criteria

- RMSE
- Akaike Information Criterion (AIC)
- R-squared

Final Indicator Set

Final Indicators

- Remittances
- Foreign reserves
- Bank credit
- Tourist receipts
- Market survey value
- Imports
- Tax revenue
- Southern Oscillation Index (El Nino)
- COVID-19 dummy
- Seasonality dummies

- Selected eight variables plus the COVID-19 and seasonality dummy variables
- Closely follow Samoa's quarterly real GDP
- All indicators have a strong and positive correlation with real GDP
- Not only correlate well with GDP but also maintain this correlation consistently over time.

2. The Three Nowcasting Models

- Evaluate and choose the model through econometric techniques to find the most suitable one;
- Three standard methods available for nowcasting:

Method	Purpose
Bridge	Aggregates monthly data to quarterly
MIDAS	Mixed-frequency regression
U-MIDAS	Flexible mixed-frequency regression

3. Back-testing

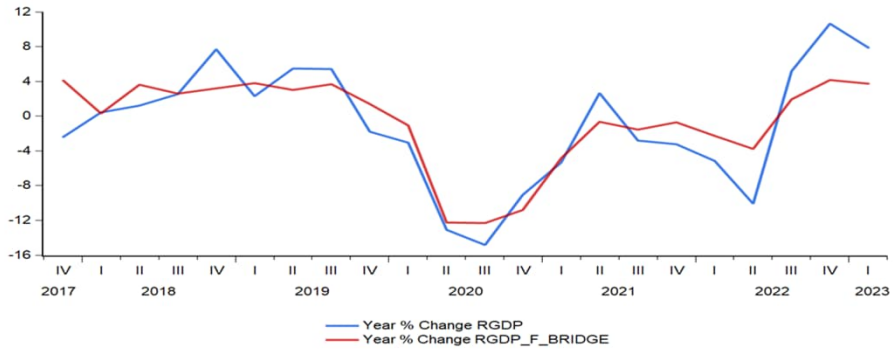
- Conducting in-sample nowcasting exercises to test the model's forecasting ability over time
- Three distinct sub-samples: full sample, pre-COVID sample, and during/post-COVID sample
- Results indicated that *combining the three forecasting methods delivers the most accurate predictions*, particularly during the COVID and post-COVID periods.

Table 5. Forecast Evaluation

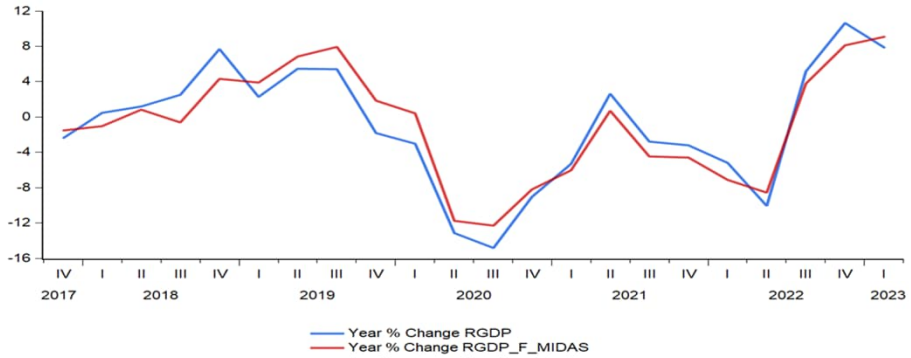
Forecast Method	Full Sample (2008Q4-2023Q1)		Pre-COVID Sample (2008Q4-2019Q4)		COVID/Post-COVID Sample (2020Q1-2023Q1)	
	RMSE	Theil U2	RMSE	Theil U2	RMSE	Theil U2
RGDP_F_FC	8.57	0.36	9.03	0.43	5.96	0.19
RGDP_F_BRIDGE	12.46	0.52	12.29	0.57	11.04	0.36
RGDP_F_MIDAS	10.26	0.44	11.04	0.53	6.49	0.21
RGDP_F_UMIDAS	10.02	0.44	11.13	0.54	4.41	0.13

Note: The lower the RMSE and Theil U2 statistics, the better the forecasting performance. Theil U2 statistics also suggests that the forecasting model has a lower sum of squared errors than the naïve forecast, indicating better performance.
Source: Authors' calculations.

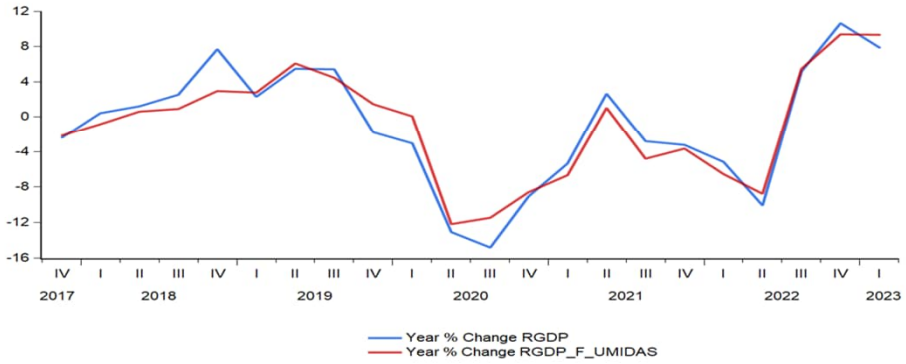
a. Bridge



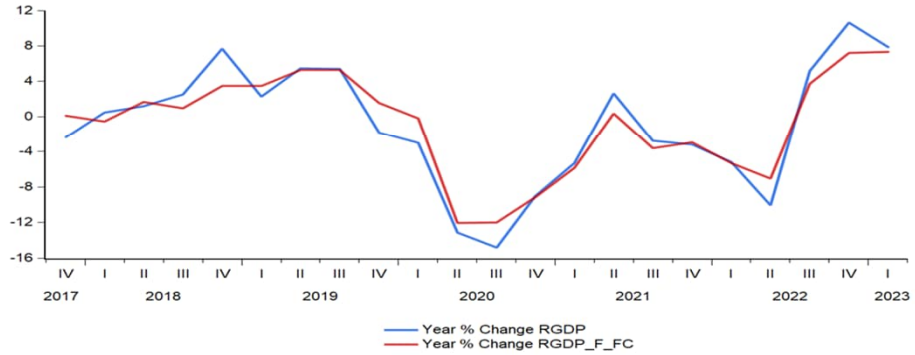
b. MIDAS



c. U-MIDAS

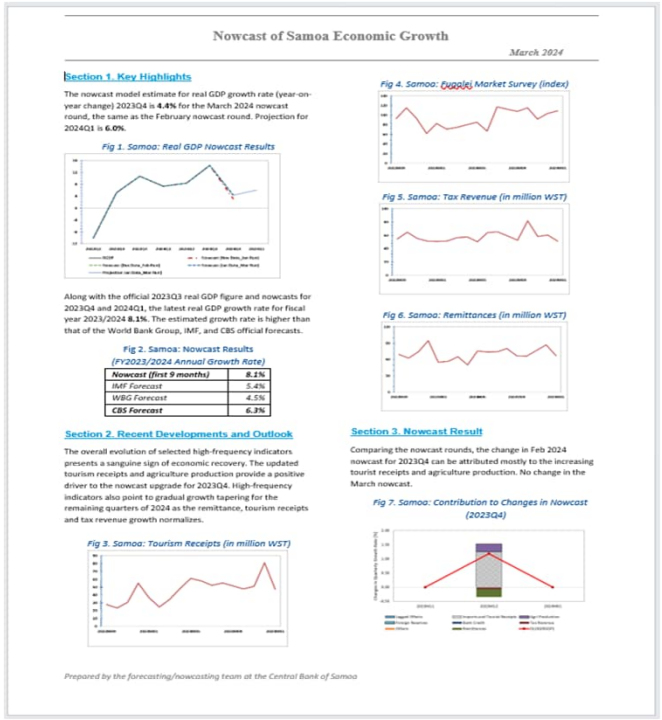


d. Forecast Combination



4. Documentation

- Documenting the nowcasting process for future reference.
  - a one-pager reporting template (CBS Board and executive management) – *ad hoc*
  - IMF Working Paper



Nowcast of Samoa Economic Growth

March 2024

Section 1. Key Highlights

The nowcast model estimate for real GDP growth rate (year-on-year change) 2023Q4 is **4.4%** for the March 2024 nowcast round, the same as the February nowcast round. Projection for 2024Q1 is **6.0%**.

Fig 1. Samoa: Real GDP Nowcast Results

Along with the official 2023Q3 real GDP figure and nowcasts for 2023Q4 and 2024Q1, the latest real GDP growth rate for fiscal year 2023/2024 **8.1%**. The estimated growth rate is higher than that of the World Bank Group, IMF, and CBS official forecasts.

Fig 2. Samoa: Nowcast Results
(FY2023/2024 Annual Growth Rate)

Nowcast (first 9 months)	***8.1%***
IMF Forecast	*5.4%*
WBG Forecast	*4.5%*
CBS Forecast	***6.3%***

Section 2. Recent Developments and Outlook

The overall evolution of selected high-frequency indicators presents a sanguine sign of economic recovery. The updated tourism receipts and agriculture production provide a positive driver to the nowcast upgrade for 2023Q4. High-frequency indicators also point to gradual growth tapering for the remaining quarters of 2024 as the remittance, tourism receipts and tax revenue growth normalizes.

Fig 3. Samoa: Tourism Receipts (in million WST)

Fig 4. Samoa: Supplier Market Survey (index)

Fig 5. Samoa: Tax Revenue (in million WST)

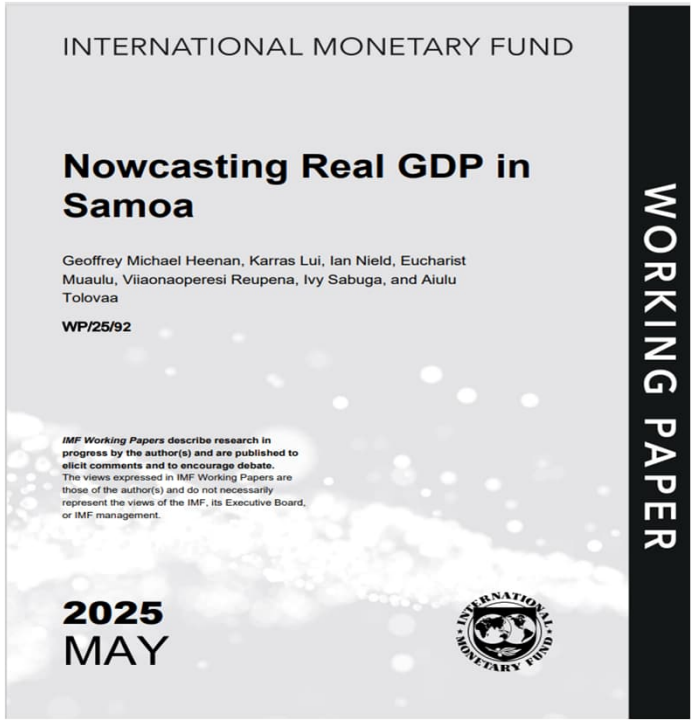
Fig 6. Samoa: Remittances (in million WST)

Section 3. Nowcast Result

Comparing the nowcast rounds, the change in Feb 2024 nowcast for 2023Q4 can be attributed mostly to the increasing tourist receipts and agriculture production. No change in the March nowcast.

Fig 7. Samoa: Contribution to Changes in Nowcast (2023Q4)

Prepared by the forecasting/nowcasting team at the Central Bank of Samoa



INTERNATIONAL MONETARY FUND

Nowcasting Real GDP in Samoa

Geoffrey Michael Heenan, Karras Lui, Ian Nield, Eucharist Muaulu, Viiaonaoperesi Reupena, Ivy Sabuga, and Aiulu Tolovaa

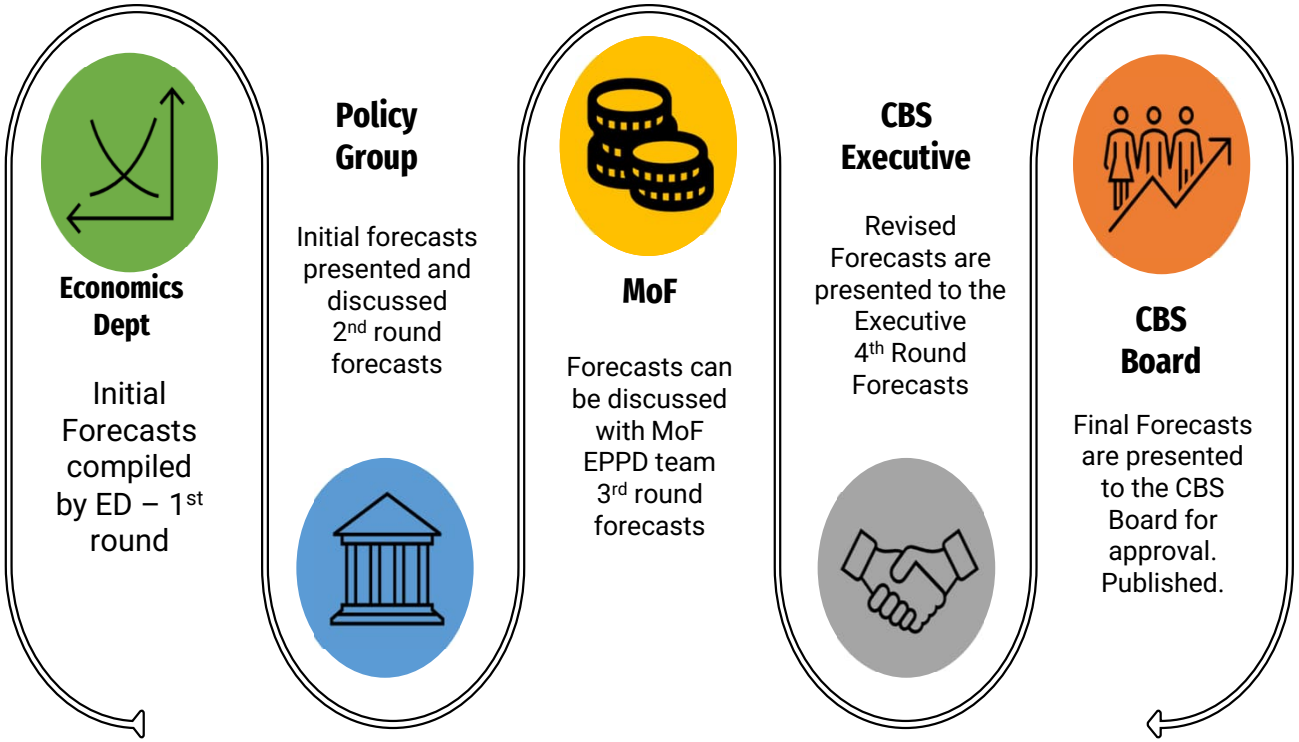
WP/25/92

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2025
MAY

WORKING PAPER

CBS processes pre-FPAS



1. Forecasts were done twice a year (June-July and Jan-Feb)
2. FPAS process (Forecast) compiled around 1-2 month ahead of BM
3. PR usually 1 week after BM
4. MPS 3-5 months after BM (due to availability)
5. All forecast models are excel based
6. Involved a lot of user expertise and consultations
7. Communications were very technical due to nature of MP

New FPAS Process

- Establish a *regular cycle* of collection, forecasting and policy analysis
- Adoption of regular MP forecasting (6 months) cycle (June and December)
- Announced calendar (2 months preparation) for the biannual monetary policy board meetings
- Formalized a
  - Monetary Policy Working Group (MPWG) (DG–Policy, ED, FMD) meet twice in a cycle and a
  - Monetary Policy Committee (MPC) – Executive + Policy Group (FSRSD, ED, FMD and FSSD) + Accounts

Revamped Communications

- *Production and streamlining of key reports* should be timed to coincide with these meetings (BM Paper, Press Release and MPS full report)
- *Layered communications approach*
  - Simplified PR for general public,
  - MPS – technical report,
  - More focused PR – for markets and Govt,
  - Infographics for social media

Revamped Communications

- New CBS Website – improve CBS communication – publish MP Framework, current MP stance, MPS reports, PRs, papers
- Develop a *CBS Communication framework* – Communications policy, trainings etc…
- New Communications Manager now onboard
- Communications TA (general) from Tautai (Australian Aid)
  - Developed communications strategy
  - Trainings on report writing, plain language, etc…

Quarterly Projections Model (QPM)

- Using EViews, looked at a canonical QPM consisting of *four main components*: the aggregate demand IS curve, the forward-looking Phillips curve, the uncovered interest parity condition (UIP), and the monetary policy rule
- We now have a fully-functional QPM in place (July 2026) together with PFTAC, STI and ICD HQ

Issues identified from FPAS Missions

Data Issues

- Frequency (monthly, quarterly, annual), Coverage (sales, land prices, tax
- Manual processes
  - Combination of Cut and paste and Excel links
  - Looking at VBA-based
- No metadata/documentation - allocate staff to data curator roles for acquiring, updating, storing and overall management of data sets
- Decentralized (to some extent) – trying to centralize
- Access rights and timely backup (drive)
- Where should it be located - Local hard disk, shared server – considering cloud based (Onedrive)
- Future conversion to another database solution (open source), APIs



Data Management

- Eliminate, or significantly reduce, the number of steps and manual data entry
- Oversee audit of critical spreadsheets and other analytical systems to
  - *streamline data management procedures*, and
  - *remove redundant or obsolete files, tables and charts*.
  - To document file locations, access, purpose and updating procedures
- Implement robust, low risk processes for *storing, controlling and retrieving data from databases* accessible by CBS staff
- Look at a Memoranda of understanding (MoU) with Government - Ministry of Customs & Revenue, Ministry of Finance, and the Samoa Bureau of Statistics (SBS) for the acquisition and dissemination of data.

Adoption of FPAS

Tools – EViews used by both ED and FMD, GUI Model,

Models

a) Nowcasting of GDP - fully incorporated into our forecasting processes and is fitted into our excel based forecast tools

b) Core inflation is basically for CBS internal reporting only. Updated monthly and presented to every board meeting.

   i. Look to forecast headline inflation

c) ERPT is completed

d) QPM is also completed

e) Data Management. Staff already carrying out basic data management in centralizing, cleaning duplicates and look to setup documentation, in and out sheets

   i. Hired 1 Forecasting economist and will hire a data management officer.

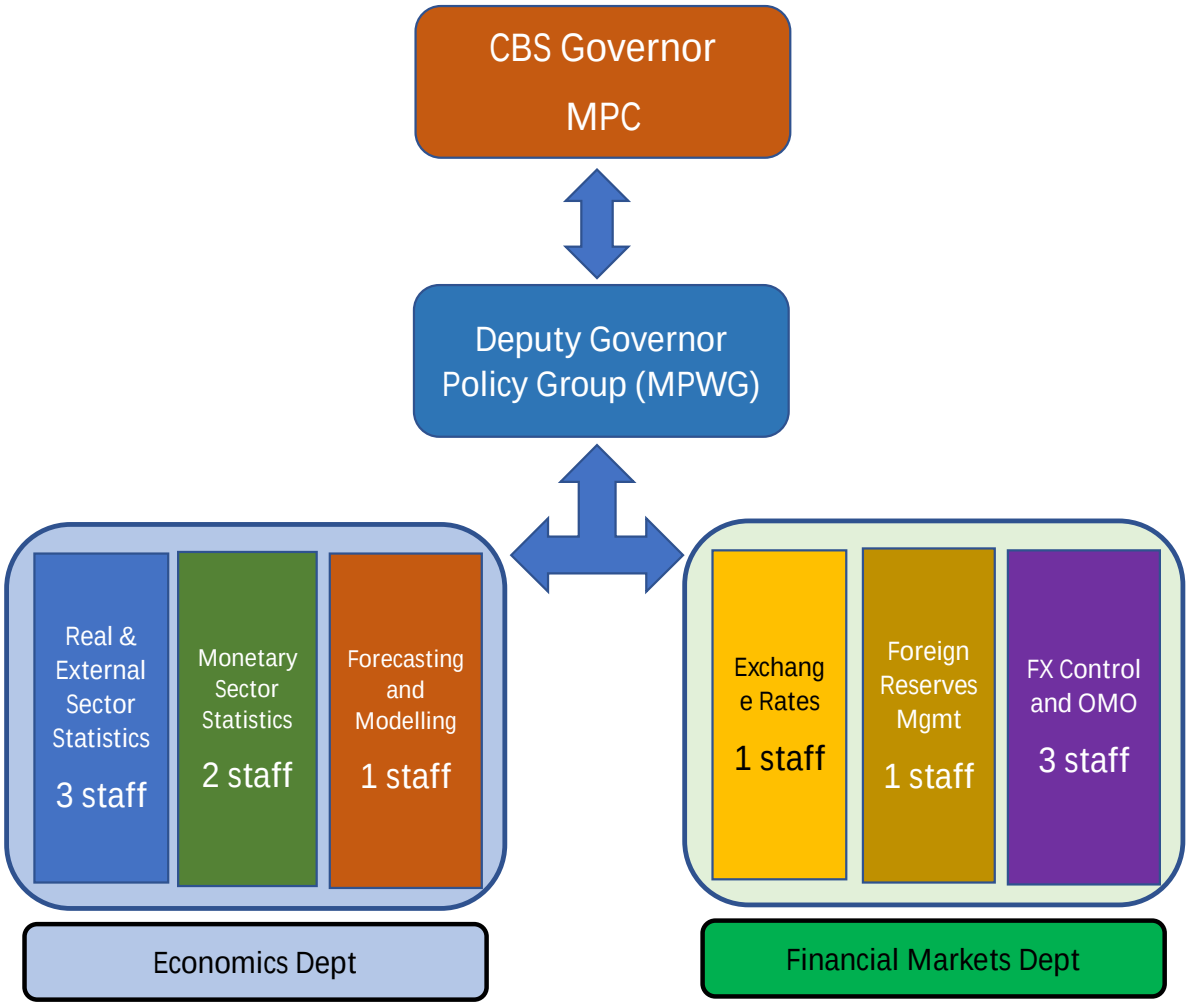
Adoption of FPAS

e) FPAS process
   i. Now managed to get the Board to agree on a set BM dates
   ii. Work backwards to set the MPC and MPWG meeting dates
   iii. Set deadlines for economist to compile their forecasts and texts for the MPS

f) Communication – MPS now published 1 week after BM whereas before 3-5 months after BM depending on free time.
   a) Simplify CBS MP reports – Plain language + Infographics

Challenges

i. Staff Capacity – econometrics, learning EViews, Matlab, Python? – Higher education (Masters Degree)
ii. Limited Staff number – ED has 6 economists and FMD has 2
    i. High risk of staff leaving CBS
iii. Other competing duties – Economists have their own primary duties – external sector, real sector, monetary sectors, exchange rates, foreign reserves
iv. Data gaps – lack of available relevant data (tax, sales, real estate prices etc…)
    a) Data sharing arrangements





i. Need to plan out new Forecasting/Modelling Unit including staffing – division of duties
ii. Data Management side/unit

Benefits/Impact of FPAS TA

i. Operationally
   a) A more *structured*, *time bound*, *focused* and *efficient* MP forecasting process from the first step right up to BM and PR
   b) *Efficiency* in data entry, storage and organization

ii. Analysis
   a) Improvement in forecast analyses – core inflation, output gap, passthrough
   b) More economics and less 'gut feeling'

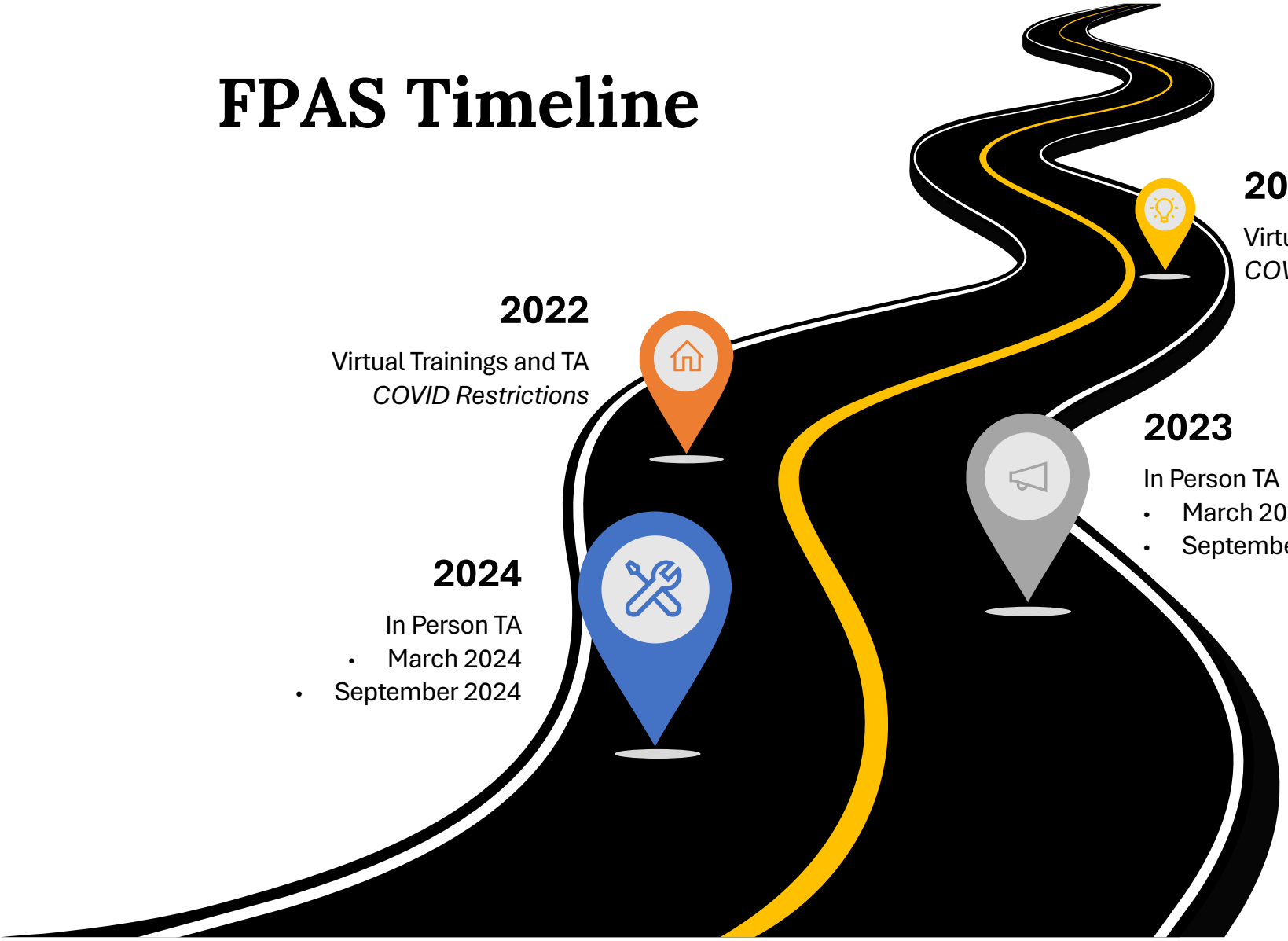
iii. Policy decision making
   a) Better linkages between forecasts and MP stance
   b) Much improved communication (targeted, better linked)

iv. Staff capacity
   a) Better trained and professional staff

Way Forward!

- Nowcasting Framework review:
  - Data update and refinement – include or exclude any new/old indicator
  - Model update – how are the latest results/performance of each approach?
- Incorporate the QPM into the new FPAS process
- Prepare the Nowcasting one-pager on a regular basis to align with BMs
- Request a TA from the IMF that focuses on Communications
- Ongoing review of CBS reports to enhance communication and outcome
- Expand the Forecasting Division within the Economics Department

FPAS Timeline



2021

Virtual Trainings and TA
COVID Restrictions

2022

Virtual Trainings and TA
COVID Restrictions

2023

In Person TA
- March 2023
- September 2023

2024

In Person TA
- March 2024
- September 2024

2025

In Person TA
- September 2025

Fa’afetai Lava!